

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	534264
Award Program	New Innovator
Project Title	Improving Simulations of Water Dynamics and Crop Yield in the Corn Belt
Grant Start Date	10/01/2017
Grant End Date	12/31/2021
Total Grant Budget	\$600,000
Total FFAR Budget	\$300,000
Matching Funder(s)	Iowa Crop Improvement Association
Final Report Date	09/12/2022
Project Director/Principal Investigator Information	
Full Name	Sotirios Archontoulis
Email	sarchont@iastate.edu
Phone	5152947413
Grantee Organization Information	
Organization Name	Iowa State University, Department of Agronomy
Address	
City, State Zip	Ames, Iowa, 50011
Tax ID	



Authorized Signing Official (ASO) Information	
Name	Tammy Polaski
ASO Title	Assoc. Director, Office of Sponsored Programs Administration
Email	Ospa-awards@iastate.edu
Phone	515-294-5225

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Iowa and US Corn Belt in general

- 1.2. How many new jobs were created because of this grant funding?

Six in total; four postdocs, and two graduate students

- 1.3. How many jobs were maintained because of this grant funding?

Each year I had 2-3 students working on the project

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Most of today's crop models lack representations of shallow groundwater, which have resulted in significant errors in simulating yield response to precipitation – an attribute of paramount importance for climate change assessments. This project addressed this deficiency by 1) generating new knowledge from field research on groundwater table dynamics across environments and their influence on root and crop processes and 2) developing new protocols to enable the simulation of groundwater in crop models at scale. Furthermore, this project developed a new web-based tool with real-time spatiotemporal maps of soil moisture in the



US Corn Belt to aid farmers' decision support (2018-2022). Project results have been published in peer-reviewed journals and disseminated at scientific conferences. Key findings include 1) when soil moisture is above 95% saturation, the increase in rooting depth ceases; 2) one-dimensional water balance models (part of crop models) can generate watertable dynamics in the US Corn Belt; 3) accounting for shallow water table in the crop model, the simulation of the grain yield response to precipitation improved from R^2 42% to 86% and changed the direction of N loss suggesting more N_2O emissions and less N leaching; 4) long term simulations, from 1984 to 2019 revealed that the water table becomes shallower (up to 8 mm/year) in parts of the US Corn Belt. Overall, our work addressed a long-standing challenge in crop modeling community, which is grain yield under-prediction in dry years and grain yield over-prediction in wet years. The new protocols and results generated in this project will stimulate new thinking to move us forward with climate change impact assessments as well as sustainability assessments using simulation crop models.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

Findings from this project have already been used by the agricultural community and will be used more in the future. In the crop modeling community, this project has stimulated new thinking on how to improve crop models and how to best set up our crop models for crop yield and sustainability predictions. I will mention two examples here. Dr. Grassini from the University of Nebraska collaborated with my team to develop a new framework to improve their model (Hybrid-maize) to forecast crop yields more accurately in the US Corn Belt by accounting for groundwater in some regions. This collaboration yielded a publication (Rizzo et al., 2018). A second example is Dr. Harrison from the University of Tasmania. Our pioneer work on adding waterlogging functions in the APSIM soybean and maize models (Pasley et al., 2020; Ebrahimi-Mollabashi et al., 2019), resulted in another collaboration to update the APSIM barley model with our new algorithms, which yielded 2-3 additional publications. These two examples show that our work is significant and of great interest to the scientific community. Furthermore, I have been invited to present to agronomists and international modeling groups about the work we did on groundwater table and waterlogging functions (see events to a later section).

Some key findings and interpretations that came out of this project include:

- Inclusion of water table in crop modeling significantly improves the simulation of crop yield response to precipitation, a very important aspect as we use crop models for climate change scenario analysis (Archontoulis et al., 2020)
- Soils/locations in the US Corn belt with shallow water tables have higher yields and lower inter-annual yield variability as compared to soils/locations without water table, which are more vulnerable to weather extremes (Archontoulis et al., 2020)
- It is easier to predict crop yields in soils/locations with shallow water tables early in the



growing season because the existence of watertable buffers any drought stress (Archontoulis et al., 2020)

- Model set-up by the user (e.g., inclusion of water table), initial conditions, and early season measurements are very important for accurate predictions of soil water, N and crop yields in this environment (Archontoulis et al., 2020).
- Crop yields are impacted by waterlogging (excess water) via multiple processes: a) reduced photosynthesis; b) delays in crop development; c) reduction in biological N fixation; d) reduction in root depth and e) plant N redistribution. Most of these mechanisms have been added to the APSIM model, which subsequently improved the simulation of soil and crop processes (Pasley et al., 2020). This work resulted in several invitations to present the algorithms to other modeling groups
- Using the updated model, Pasley et al. (2021) performed a scenario study for the US Corn belt and found that rotating crops reduce the risk and rate of nitrate loss.
- Vertical root growth is inhibited when volumetric soil moisture approaches 3% of saturation (Ebrahimi-Mollabashi et al. 2019). The plant roots quickly recover and keep growing once soil moisture drops to lower levels. This work resulted in a major model upgrade and the new code was released in public through APSIM's open source system.
- Maize root distributions are strongly associated with water tables (Nichols et al., 2019). We performed a large experimental effort and examined many factors that can affect root distribution (soil type, drainage condition, soil carbon, texture, etc.) and the water table depth was the No1.
- As part of our experimental work, we identified that insufficient and excessive N fertilizer input reduces maize root mass across soil types (Ordonez et al., 2021). This finding is very important and suggests that there are different optimums for crop yield and root mass production (typically the optimum N input to maximize root mass is less than that of the N input to maximize grain yield). This finding generated new thinking among scientists working on sustainability.
- Inclusion of groundwater table improves also the predictive capacity of machine learning models at the regional scale (Shahhosseini et al., 2021). Within 1.5 years of the publication of our finding, the paper has reached 60 citations, which demonstrates that the scientific community is using our results.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and



- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

First and most important, there was very little awareness of the importance of groundwater for crop production in the US Corn Belt before this project. Since the beginning of this project, the role of groundwater has been increasingly discussed and acknowledged in the scientific community as evidenced by publications mentioning groundwater table and my invitations to present in scientific conferences and workshops (see below). This project changed the way of thinking for many scientists regarding the agricultural system. I see that from my day-to-day interactions with soil and environmental scientists. The project findings have also generated significant new knowledge (see papers in section 3.1). Through this project and experimental work, we provided new data and knowledge on crop roots (see publications), updating the very old scientific information, which will help decision-making and most importantly understanding of the complex agronomic system. Our modeling work inspired other groups to work on waterlogging issues of various crops (e.g., barley) and others to think about how to improve their crop models for use in climate change research. Recently, the modeling team (AGMIP) created and distributed a survey among the global modeling community to form a global team to work further on improving waterlogging in wheat crop models. The work that was done on this project motivated the creation of this new AgMIP initiative. Furthermore, improved capacity of the APSIM model to predict crop yields and soil N dynamics (see Archontoulis et al., 2020 publication) by accounting for groundwater table and waterlogging, was a key component for securing funds from Iowa legislation to create the new "Iowa Nitrogen Initiative", which will provide stakeholders with robust N rate recommendations to improve productivity, profitability, and environmental sustainability

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

There were three goals, 1) perform experiments to generate new knowledge, 2) improve crop model, and 3) deploy the modeling system to assist decision making (see details in the proposal). All goals were achieved. The project was for 3 years but because of covid-19 it was expanded to 4.5 years. Overall, we accomplished a lot in the project (see publication list) and inspired many others to start working on this issue (groundwater and waterlogging). While the project officially ended, we are still working on two publications (see section 3.1), one is under review and the other one to be submitted. The one that is to be submitted is a high impact publication (target journal is PNAS).

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)



limit)

n/a

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Portions of the requested information is provided in other sections of this report, and I will refer to the scientific publication for the technical aspects (see section 3.1)

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

For the experimental part (goal 1), we collected data from my replications and my different environments to ensure our results are robust and hold across environments. For the modeling part (goal 2), we retrieved additional data from the literature (in addition to data generated from goal 1) to test our modeling system and ensure that the model is robust and accurate. For the application part of our modeling system for decision support at scale (goal 3), we qualitatively compared our model predictions at the US Corn Belt scale with other tools (e.g., US Drought Monitor) to ensure that both tools are directionally correct. Comparing the absolute results is challenging because the US Drought Monitor provides categorical results.

- 2.8. List key stakeholders that could be served by results of this project.

Everyone involved in maize and soybean production in the US Corn Belt, including farmers, researchers, faculty, students, industry and other state/federal organizations

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not



designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report, see information products and list of presentation in subsequent sections; this was a research project.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Covid-19 was a significant challenge and it caused delays in hiring personnel to work on this project (postdocs) as well as training them on modeling. A second and more general challenge that should be addressed in the future is the lack of groundwater monitoring systems (in agriculture land) to support crop model activities. In the project, we set up new wells and accumulated information to accomplish the stated objectives but for the future a more robust solution should be developed.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Postdocs and grad students involved in this project were given many professional development opportunities. These include travel and presentations in scientific meetings, participation in modeling meetings with international colleagues, and finally opportunities to serve as co-teaching in crop modeling training workshops

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

4 postdocs, 2 graduate students, 3 undergraduate students

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Peer-reviewed journal publications

1. Archontoulis SV, Castellano MJ, Licht MA, [Nichols V](#), [Baum M](#), [Huber I](#), [Martinez-Feria R](#), [Puntel L](#), [Ordóñez RA](#), Iqbal J, [Wright EE](#), [Dietzel RN](#), Helmers M, Vanloocke A, Liebman M, Hatfield JL, Herzmann D, Cordova SC, Edmonds P, Togliatti K, Kessler A, Danalatos G, Pasley H, Pederson C, Lamkey KR, 2020. Predicting Crop Yields and Soil-Plant Nitrogen Dynamics in the US Corn Belt. **Crop Science**, 60: 721–738. (citations = 57)
2. [Ebrahimi-Mollabashi E](#), Huth NI, Holzwoth DP, [Ordóñez RS](#), Hatfield JL, [Huber I](#), Castellano MJ, Archontoulis SV, 2019. Enhancing APSIM to simulate excessive moisture effects on root growth. **Field Crops Research** 236: 58–67. (citations = 37)
3. [Ordóñez R](#), Castellano M, Danalatos GN, [Wright E](#), Hatfield J, Burras L, Archontoulis S, 2021. Insufficient and excessive N fertilizer input reduces maize root mass across soil types. **Field Crops Research**, <https://doi.org/10.1016/j.fcr.2021.108142> (citations = 9)
4. [Ordóñez R](#), Archontoulis SV, [Martinez-Feria R](#), Hatfield J, [Wright E](#), Castellano M, 2020. Root to shoot and carbon to nitrogen ratios of maize and soybean crops in the US Midwest. **European J of Agronomy** 120, 126–130. (citations = 24)
5. [Ordóñez R](#), Castellano M, Hatfield J, Helmers M, Licht M, Liebman M, [Dietzel R](#), [Martinez-Feria R](#), [Iqbal J](#), [Puntel L](#), [Cordova C](#), [Togliatti K](#), [Wright E](#), Archontoulis SV, 2018. Maize and soybean root front velocity and maximum depth in Iowa, USA. **Field Crops Research** 215: 122–131 (citations = 59)
6. [Ordóñez R](#), Castellano M, Hatfield J, Licht M, [Wright E](#), Archontoulis SV, 2018. A solution for sampling position errors in maize and soybean root mass and length estimates. **European J of Agronomy** 96: 156–162. (citations = 11)
7. [Pasley HR](#), [Huber I](#), Castellano MJ, Archontoulis SV, 2020. Modeling flood-induced stress in soybeans. **Frontiers Plant Science** 11:62, [doi:10.3389/fpls.2020.00062](https://doi.org/10.3389/fpls.2020.00062) (citations = 22)
8. [Pasley H](#), [Nichols V](#), Castellano M, Helmers M, [Baum M](#), Klavivko E, Archontoulis S, 2021. Rotating Maize Reduces the Risk and Rate of Nitrate Leaching. **Environmental Research Letters**, <https://doi.org/10.1088/1748-9326/abef8f> (citations = 6)
9. [Nichols V](#), [Ordóñez A](#), [Wright E](#), Castellano M, Liebman M, Hatfield J, Helmers M, Archontoulis SV, 2019. Maize root distributions strongly associated with water tables in Iowa, USA. **Plant Soil**, 444: 225–238 (citations = 18)
10. Rizzo G, Edreira JR, Archontoulis SV, Yang H, Grassini P, 2018. Do shallow water tables contribute to high and stable maize yields in the US Corn Belt? **Global Food Security** 18: 27–34. (citations = 25)
11. Castellano M, Archontoulis S, Helmers M, Poffenbarger H, Six J, 2019. Sustainable intensification of agricultural drainage. **Nature Sustainability** 2, 914–921 (citations = 48)

12. [Elli EF](#), [Ciampitti IA](#), [Castellano MJ](#), [Purcell LC](#), [Naeve S](#), [Grassini P](#), [La Menza NC](#), [Moro Rosso L](#), [de Borja Reis AF](#), [Kovács P](#) and [Archontoulis SV](#) (2022). Climate Change and Management Impacts on Soybean N Fixation, Soil N Mineralization, N₂O Emissions, and Seed Yield. **Front. Plant Sci.** 13:849896. doi: 10.3389/fpls.2022.849896
13. [Baum M](#), [Licht M](#), [Huber I](#), [Archontoulis SV](#), 2020. Impacts of climate change on the optimum planting date of different maize cultivars in the central US Corn Belt. **European J Agronomy** 119, 126101. (citations = 33)
14. [Martinez-Feria R](#), [Nichols V](#), [Basso B](#), [Archontoulis S](#), 2019. Can multi-strategy management stabilize nitrate leaching under increasing rainfall? **Environmental Research Letters** 14, 124079. (citations = 14)
15. [Martinez-Feria R](#); [Castellano M](#); [Dietzel R](#); [Helmers M](#); [Liebman M](#); [Huber I](#); [Archontoulis SV](#), 2018. Linking crop- and soil-based approaches to evaluate system nitrogen-use efficiency and tradeoffs. **Agriculture, Ecosystems and Environment** 256: 131-143 (citations = 56)
16. [Puntel L](#), [Sawyer J](#), [Barker](#), [Thorburn P](#), [Castellano M](#), [Moore K](#), [Vanloocke A](#), [Heaton E](#), [Archontoulis SV](#), 2018. A systems modeling approach to forecast corn economic optimum nitrogen rate. **Frontiers in Plant Science**, 9:436, doi: 10.3389/fpls.2018.00436 (citations = 42)
17. [Shahhosseini M](#), [Hu G](#), [Huber I](#), [Archontoulis S](#), 2021. Coupling Machine Learning and Crop Modeling Improves Crop Yield Prediction in the US Corn Belt, **Nature Scientific Reports**, 11:1606 (citations = 59)
18. [Shahhosseini M](#), [Martinez-Feria R](#), [Hu G](#), [Archontoulis SV](#), 2019. Maize yield and nitrate loss prediction with machine learning algorithms. **Environmental Research Letters**, 14, 124026 <https://doi.org/10.1088/1748-9326/ab5268> (citations = 82)
19. [Couëdel A](#), [Edreira JIR](#), [Lollato RP](#), [Archontoulis SV](#), [Sadras V](#), [Grassini P](#), 2021. Assessing environment types for maize, soybean, and wheat in the United States as determined by spatio-temporal variation in drought and heat stress. **Agricultural and Forest Meteorology**, 307: 108513 (citations = 10)
20. [Kimball B](#), [Boote KJ](#), [Hatfield J](#), [Ahuja L](#), [Stockle C](#), [Archontoulis S](#), [Baron C](#), [Basso B](#), [Bertuzzi P](#), [Constantin J](#), [Deryng D](#), [Dumont B](#), [Durand J-L](#), [Ewert F](#), [Gaiser T](#), [Gayler S](#), [Hoffmann M](#), [Jiang Q](#), [Kim S-Q](#); [Lizaso J](#), [Moulin S](#); [Nendel C](#), [Parker P](#), [Palosuo T](#), [Priesack E](#), [Qi Z](#), [Amit S](#), [Stella T](#), [Tao F](#), [Thorp K](#), [Timlin D](#), [Twine T](#), [Webber H](#), [Willaume M](#), [Williams K](#), 2019. Simulation of Maize Evapotranspiration: An Inter-comparison among 29 Maize Models. **Agricultural and Forests Meteorology** 271, 264–284. (citations = 46)
21. [Hao J](#), [Chai YN](#), [Lopes LD](#), [Ordóñez R](#), [Wright E](#), [Archontoulis S](#), [Schachtman D](#), 2021. The effects of soil depth on the structure of microbial communities in agricultural soils in Iowa, USA. **Applied Environmental Microbiology**, 87:e02673-20. <https://doi.org/10.1128/AEM.02673-20>. (citations = 26)

Journal articles under review / in preparation

1. [Archontoulis et al](#). The undervalued groundwater has enormous consequences on productivity and sustainability evaluations – towards improving simulation models. In preparation for submission to PNAS.
2. [Elli E](#), [Archontoulis](#), 2022. Dissecting the contribution of weather and management on

water table dynamics under present and future climate scenarios in the US Corn Belt.
Agronomy for Sustainable Development journal (under review)

Newsletters

1. Archontoulis S, Licht M, 2022. [Early May Soil Conditions: 2022 Benchmark](#). ICM blog news, Iowa State University.
2. Archontoulis S, Licht M, Castellano M, 2021. [FACTS soil moisture benchmarking tool](#). ICM blog news, Iowa State University
3. Archontoulis S, Licht M, Castellano M, 2021. [Changes to Forecasting and Assessment of Cropping sysTemS website \(FACTS\)](#). ICM blog news, Iowa State University
4. Archontoulis S, 2021. [Soil N Benchmark April 19, 2021](#). ICM blog news, Iowa State University
5. Archontoulis S, Licht M, 2021. [Soil moisture Benchmark April 19, 2021](#). ICM blog news, Iowa State University
6. Archontoulis S, Licht M, Clemens Z, 2020. [FACTS update: Iowa soils drier, not a sign of concern](#). ICM blog news, Iowa State University
7. Archontoulis S, Licht M, Clemens Z, 2020. [FACTS Report: N Mineralization Low, But Will Rise](#). ICM blog news, Iowa State University
8. Archontoulis S, Licht M, 2019. [Update on crop and soil status across the I-States – August 21, 2019](#). ICM blog news, Iowa State University
9. Archontoulis S, Licht M, 2019. [New regional scale feature added to FACTS](#). ICM blog news, Iowa State University
10. Archontoulis S, Licht M, 2019. [FACTS regional scale results as of June 19, 2019](#). ICM blog news, Iowa State University
11. Archontoulis S, Licht M, 2018. [Updated FACTS forecast](#). ICM blog news, Iowa State University
12. Archontoulis S, Licht M, 2018. [FACTS releases first 2018 yield forecast](#). ICM blog news, Iowa State University
13. Archontoulis S, Castellano M, Licht M, 2018. [Soil water, residue, and nitrogen status entering the 2018 growing season](#). ICM blog news, Iowa State University
14. Archontoulis S, Licht M, Castellano M, 2018. [High yields despite drought conditions](#). ICM blog news, Iowa State University

Decision support web-tools

1. <https://crops.extension.iastate.edu/facts/soil-moisture-15cm>
2. <https://crops.extension.iastate.edu/facts/soil-moisture-45cm>
3. <https://crops.extension.iastate.edu/facts/soil-moisture-100cm>
4. <https://crops.extension.iastate.edu/facts/soil-n-mineralization>

- 3.2. Please provide a list of citations for the information products produced during the project.

The citations for the published journal articles are provided in section 3.1 (source: google scholar). The total citations as of July 2022 are 684 and increasing.

Invited presentations

1. Archontoulis S, 2022. Biophysical modeling of cropping systems. Making Climate Smart Ag Work. Zoom Conference organized by **AgMIP** May 24-26, 2022.
2. Archontoulis S, 2021. Water table impacts on productivity and sustainability. **University of Nebraska Lincoln**, Department of Agronomy seminar, Dec 10, 2021.
3. Archontoulis S, 2021. Tracking soil conditions and predicting crop yields. **Agronomy Society of America**, Symposium on AgMIP/NASA Harvest Activities to advance use of satellite remote sensing for crop modeling and monitoring for food security, Salt Lake City, Utah November 10, 2021.
4. Archontoulis S, 2021. Water table impacts on productivity and sustainability. **XXIX Aapresid Conference**, Argentina, e-presentation, August 17, 2021.
5. Archontoulis S, 2020. Experiments, Data, and Crop modelling. Department Seminar, **University of Kentucky**, Department of Plant and Soil Sciences, e-presentation, October 2, 2020.
6. Archontoulis S, 2020. Modeling waterlogging effects on corn and soybean with APSIM. Workshop on extreme weather. **Montpellier, France** Feb 7, 2020.
7. Archontoulis S, 2020. Modeling shallow water tables impacts on productivity and soil health in the US Midwest. Workshop organized by FFAR on linking soil and watershed health to water management. Erickson Alumni Center, **West Virginia University**, Jan 23, 2020.
8. Archontoulis S, 2019. Overcoming challenges in modelling interactions between soil carbon, water balance, and nitrogen cycling in the US Corn Belt. **Agronomy Society of America, Symposium on North American Agriculture: Need for Sustainability and Resilience**. San Antonio, Texas, November 13th, 2019.
9. Archontoulis S, 2019. Predicting crop yields and soil-plant N dynamics in the US Corn Belt with APSIM. **Agronomy Society of America, Symposium on Predictive Agriculture Foundations**, San Antonio, Texas, November 11th, 2019.
10. Archontoulis S, 2019. Coupling machine learning with crop modelling to improve prediction in agriculture. **ARS-USDA** leadership meeting, Beltsville, MD, September 20th, 2019.
11. Archontoulis S, 2019. Improving simulation of soil water and crop yields in the US Midwest. **FFAR**, Roster our future conference, Washington DC, February 5th, 2019.
12. Archontoulis S, 2019. Cropping systems research in the US Midwest. **CSIRO**, Research seminar, Toowoomba, Australia, March 4th, 2019.
13. Archontoulis S, 2018. Shallow groundwater table impacts on crop production and environment sustainability. **University of Iowa**, Department seminar, Iowa City, January 26, 2018.
14. Archontoulis S, 2020. *Crop roots and soil water*. **Indiana Certified Crop Adviser (CCA) Conference**, Indianapolis, IN, December 2020.
15. Archontoulis, Castellano, 2019. *Decision Support Tools: the ISU cropping systems toolbox*. **INREC (Iowa Nutrient Research & Education Council)** Huxley, Iowa, December 9, 2019.

16. Archontoulis S, 2019. *Crop modelling of high water table situations*. **19th Annual Drainage Research Forum, Brooking, South Dakota**, November 26, 2019.
17. Archontoulis S, 2019. *Measurements and simulations of corn and soybean roots in Iowa*. **Pioneer North America Agronomy Conference**, Ames, IA, January 29, 2019.

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Please see section 3.1, peer reviewed articles published

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

1. <https://crops.extension.iastate.edu/facts/soil-moisture-15cm>
2. <https://crops.extension.iastate.edu/facts/soil-moisture-45cm>
3. <https://crops.extension.iastate.edu/facts/soil-moisture-100cm>
4. <https://crops.extension.iastate.edu/facts/soil-n-mineralization>

The above webpages provide soil moisture information (including stress threshold) at different soil depths for the entire US Corn Belt. The data displayed as generated by the improved APSIM model, which considers groundwater table dynamics in the landscape and subsurface drainage. The webpages are updated every week.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	



n/a

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

n/a

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

Our model improvements (addition of excess moisture stress on root growth) have been incorporated in the official APSIM model release and through APSIM open source release systems have been distributed in public. Website = <https://www.apsim.info/>

We published in peer-reviewed journals nearly 95% of our research findings while we have two more in the process of publication (one paper under review and one to be submitted). The published journal articles provide both the research data and interpretations of the findings.

We created a free, public website and provided regional scale maps, in which users and decision makers can easily access the information. Website = <https://crops.extension.iastate.edu/facts/>

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Yes, we generated new data via

1. Field measurements
2. Synthesis of literature papers and USDA NASS data surveys
3. Simulations

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Yes, all the data are connected via modeling, some data are used to drive the model (weather,



soil) and some measurements used to compare model simulations with observations

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Most of the data generated in this project is already published except two papers that are in the process. The data and metadata are included in the published papers (copies provided). In the attached zip file, we include the publications and additional data/metadata together with a file inventory.

As a final comment, I want to say that the New Innovator FFAR award has shaped my career path in a very positive way. Always grateful to FFAR for this award.

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Sotirios Archontoulis

PI Signature:  Date: 9/21/2022

Authorized Signing Official (ASO) Name: Tammy Polaski, Assoc. Director,

ASO Signature:  Date: 9/23/2022